EMPLOYEE ATTENDANCE TRACKER

WEB PROGRAMMING PROJECT

A MINI-PROJECT REPORT

Submitted by

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IN

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RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

An Autonomous Institution
CHENNAI

BONAFIDE CERTIFICATE

Certified that the project entitled “EMPLOYEE ATTENDANCE TRACKER”, submitted as a CS23531 – Web Programming Project, is the bonafide work of **MADHAV S - (2116241001130)**, who carried out the work under my supervision.

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INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

Accurate attendance records are a basic requirement for every organisation. They support payroll, shift planning, leave management and the monitoring of punctuality. Paper registers and spreadsheets are slow to maintain, easy to alter and difficult to summarise. This project presents a web-based Employee Attendance Tracker that replaces them with a single, lightweight and responsive application. Users sign in through a login page, and the system provides separate screens for managers and employees. Employees can check in and check out with one click and see a live timer of the time worked. Managers can review the daily attendance of the whole team, mark leave or absence, correct check-in and check-out times, and add or remove employees.

The application is built with HTML for structure, CSS for styling and JavaScript for behaviour. Passwords are never stored in plain text: they are hashed with SHA-256 using the Web Crypto API. Attendance status (present, late, on leave, absent or not in yet) is calculated automatically from the recorded times, and an arrival after 09:30 is marked late. A dashboard with summary cards, a status chart, a seven-day trend chart and a list of employees who are in gives managers a quick view of the day. A monthly report shows present, late, leave and absent days, hours worked and an attendance percentage for each employee, and can be copied as CSV. Light and dark themes, smooth transitions and a mobile-friendly layout improve usability across devices. No installation, server or database is needed, so the application can be hosted free of cost on platforms such as GitHub Pages.

By combining a clean interface with simple, transparent attendance rules, the system aims to be fast, accurate and easy to use. The proposed system also provides a foundation for future improvements such as a server-side database with secure authentication, leave requests and approvals, QR-code or biometric check-in, e-mail notifications, report export to PDF or Excel, and integration with payroll software.

ACKNOWLEDGEMENT

First, we thank the almighty God for the successful completion of the project. Our sincere thanks to our chairman **Mr. S. Meganathan, B.E., F.I.E.,** for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson **Dr. Thangam Meganathan,** for her enthusiastic motivation which inspired us a lot in completing this project and Vice-Chairman **Mr. Abhay Shankar Meganathan, B.E., M.S.,** for providing us with the requisite infrastructure.

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MADHAV S

Contents

1 INTRODUCTION 7

1.1 INTRODUCTION 7

1.2 SCOPE OF THE WORK 7

1.3 PROBLEM STATEMENT 7

1.4 AIM AND OBJECTIVES OF THE PROJECT 7

2 SYSTEM SPECIFICATIONS 8

2.1 HARDWARE SPECIFICATIONS 8

2.2 SOFTWARE SPECIFICATIONS 8

3 MODULE DESCRIPTION 9

3.1 LOGIN AND AUTHENTICATION MODULE 9

3.2 DASHBOARD MODULE 9

3.3 ATTENDANCE MANAGEMENT MODULE 9

3.4 MONTHLY REPORT MODULE 9

3.5 TEAM MANAGEMENT MODULE 9

3.6 EMPLOYEE SELF-SERVICE MODULE 9

3.7 THEME AND RESPONSIVE UI MODULE 10

4 SCREENSHOTS 11

5 CONCLUSION AND FUTURE ENHANCEMENT 17

REFERENCES 17

List of Figures

4.1 Login page 11

4.2 Manager dashboard 12

4.3 Daily attendance management 12

4.4 Monthly attendance report 13

4.5 Team management 13

4.6 Employee check-in screen 14

4.7 Employee checked in with live timer and calendar 14

4.8 Dark theme 15

4.9 Mobile responsive view 16

CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION

The Employee Attendance Tracker is a web application that records and reports the daily attendance of the employees of an organisation. Employees sign in with their own username and password and mark their arrival and departure with a single button. Managers sign in to a separate view where they can see who is present, who is late and who is on leave, correct entries, manage the list of employees and study monthly reports. The application runs entirely in the browser and requires no installation.

1.2 SCOPE OF THE WORK

The system supports secure login with two roles (manager and employee), one-click check-in and check-out, automatic late detection, manual marking of leave and absence, a dashboard with charts, a searchable daily attendance table with date selection, a monthly report with attendance percentages, employee management, a light and dark theme, and a responsive layout for desktops, tablets and phones. It is intended for small offices, schools, laboratories, workshops and student projects where a simple and free attendance system is needed.

1.3 PROBLEM STATEMENT

Many small organisations still record attendance in paper registers or shared spreadsheets. These methods are slow, prone to mistakes and open to manipulation, and monthly totals must be calculated by hand. Commercial attendance systems solve these problems but are often expensive, need special hardware or require complex set-up. A free, simple and accurate web application that provides login, check-in and check-out, automatic status calculation and ready-made reports can save time and improve the reliability of attendance records.

1.4 AIM AND OBJECTIVES OF THE PROJECT

The project aims to design and develop an accurate, responsive and user-friendly web application for tracking employee attendance. The objectives are:

- to provide a secure login page with separate manager and employee views;
- to let employees check in and check out and view their own history;
- to calculate present, late, leave and absent status automatically;
- to give managers a dashboard, daily table, monthly reports and employee management; and
- to provide a responsive light and dark interface that can be hosted free on GitHub Pages.

CHAPTER 2
SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	Intel i5
Memory	8 GB minimum
Storage	1 TB minimum

2.2 SOFTWARE SPECIFICATIONS

Operating system	Windows 10
Front-end	HTML5, CSS3, JavaScript (ES6)
Web browser	Google Chrome / Microsoft Edge / Firefox
Code editor	Visual Studio Code
Version control and hosting	Git and GitHub Pages
Data storage	Browser localStorage (no database server required)
Security	SHA-256 password hashing with the Web Crypto API

CHAPTER 3
MODULE DESCRIPTION

3.1 LOGIN AND AUTHENTICATION MODULE

This module controls access to the application. The user enters a username and a password, and the password can be shown or hidden with a toggle. The entered password is converted to a SHA-256 hash with the Web Crypto API and compared with the stored hash, so plain-text passwords are never saved. A wrong username or password shakes the form and displays a clear error message. After a successful login the user is taken to the manager view or the employee view according to the assigned role, the session is remembered in the browser, and the Sign out button ends it. Demo accounts are provided on the login page for quick testing.

3.2 DASHBOARD MODULE

The dashboard is the manager's landing page. Five summary cards show the number of team members and the number of employees who are in the office, late, on leave and absent, with an animated count-up. A donut chart shows the status of the team today, and a bar chart shows the attendance rate for the last seven working days. A “Who's in” list shows each employee who has checked in with the check-in time and status.

3.3 ATTENDANCE MANAGEMENT MODULE

This module lists every employee for a selected date, which can be changed with the date picker, and a search box filters the list by name or department. For the current day the manager can check an employee in or out; for any day the manager can mark leave or absence, edit the in and out times directly, or reset the entry. The status is calculated automatically: an arrival after 09:30 is late, a past weekday with no entry is absent, and the hours worked are calculated from the in and out times. Each action shows a short confirmation message.

3.4 MONTHLY REPORT MODULE

For any selected month this module counts the present, late, leave and absent days of each employee, adds up the hours worked, and shows an attendance meter. Attendance is calculated as days attended divided by days attended plus days absent; leave days, weekends and days before an employee joined are not counted against the employee. The report can be copied to the clipboard as CSV and pasted into a spreadsheet.

3.5 TEAM MANAGEMENT MODULE

The manager adds an employee by entering a name, department, a unique username and a password of at least six characters. The password is hashed before it is saved. An employee is removed with a two-step confirmation (the button changes to “Click again to remove”), which also deletes that employee's attendance history.

3.6 EMPLOYEE SELF-SERVICE MODULE

After login an employee sees a personal page with the current status, a large Check in / Check out button and a live timer of the time on the clock. If the manager has marked the employee on leave or absent, the button is disabled and a message explains why. Below the button the employee sees the days attended, late arrivals, leave days and hours for the month, and a calendar in which every day is coloured by status.

3.7 THEME AND RESPONSIVE UI MODULE

The interface uses CSS variables so that a light and a dark theme can be switched with a single button; the system preference is used by default and the choice is remembered. Flexbox and CSS grid layouts, together with a media query for small screens, move the sidebar to the top of the page on phones so that all controls remain easy to use. Smooth transitions are used for page changes, buttons, charts and notifications, and they are disabled for users who prefer reduced motion.

CHAPTER 4
SCREENSHOTS

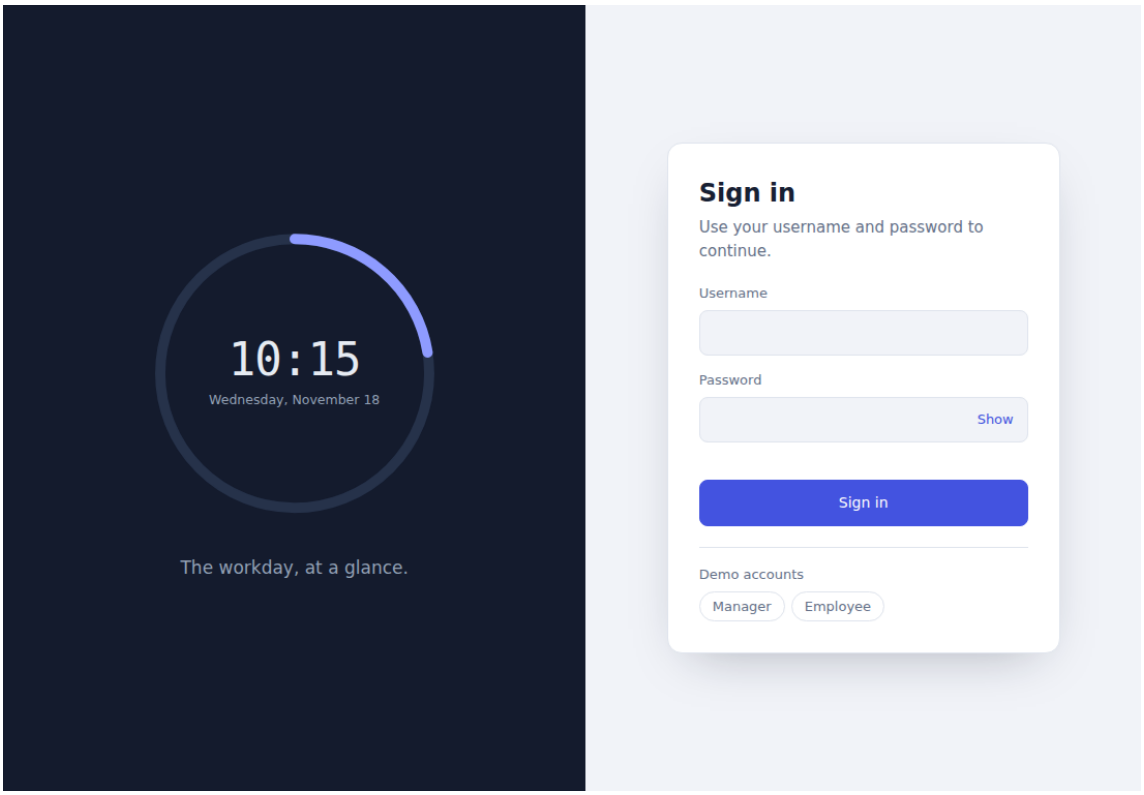


Figure 4.1: Login page

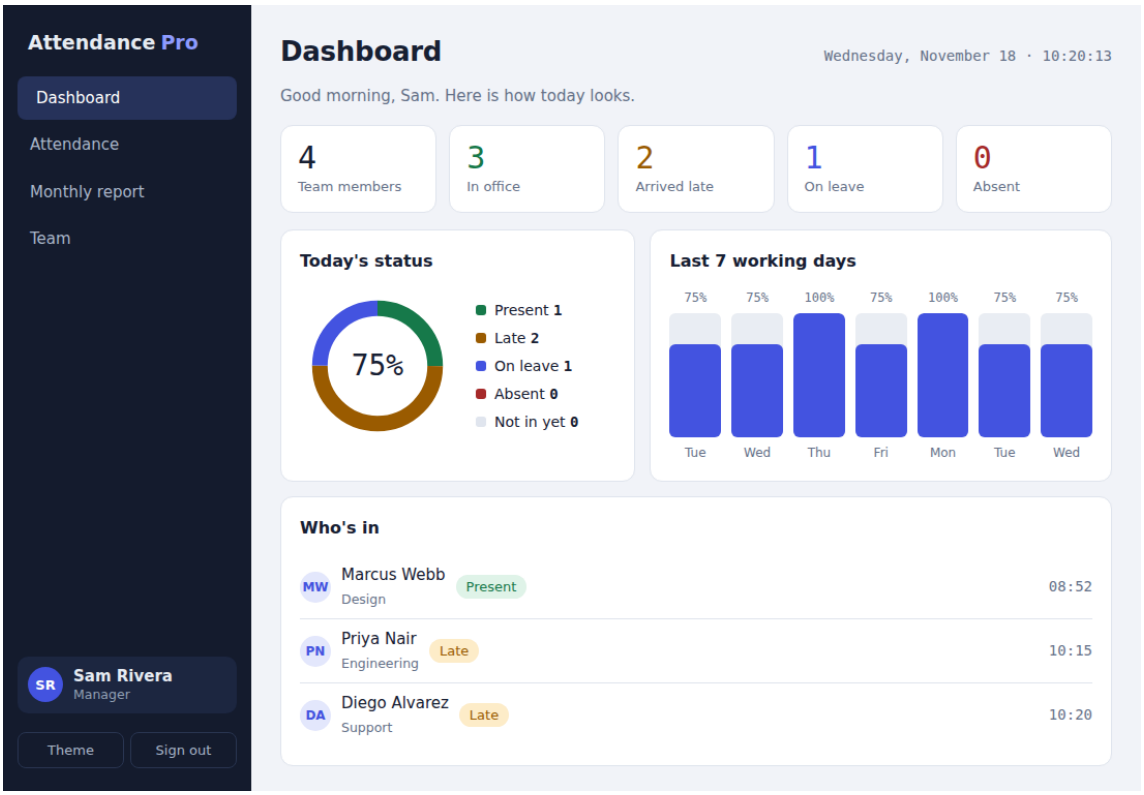


Figure 4.2: Manager dashboard

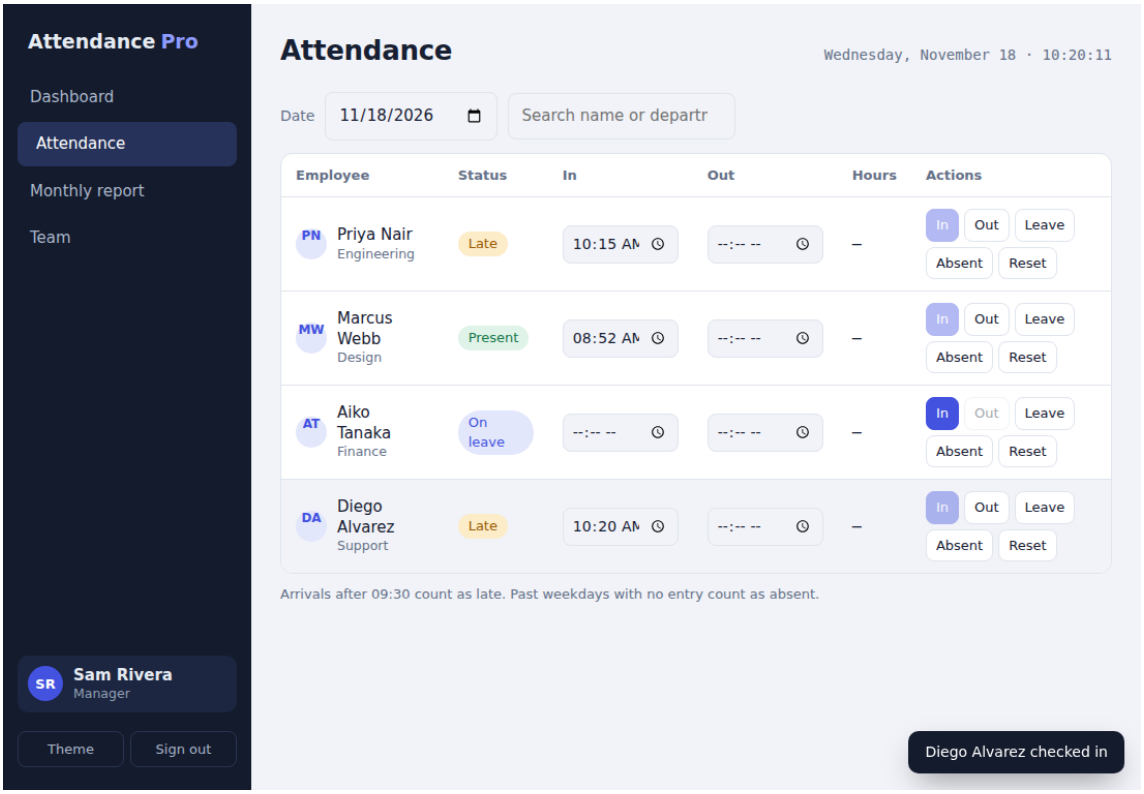


Figure 4.3: Daily attendance management

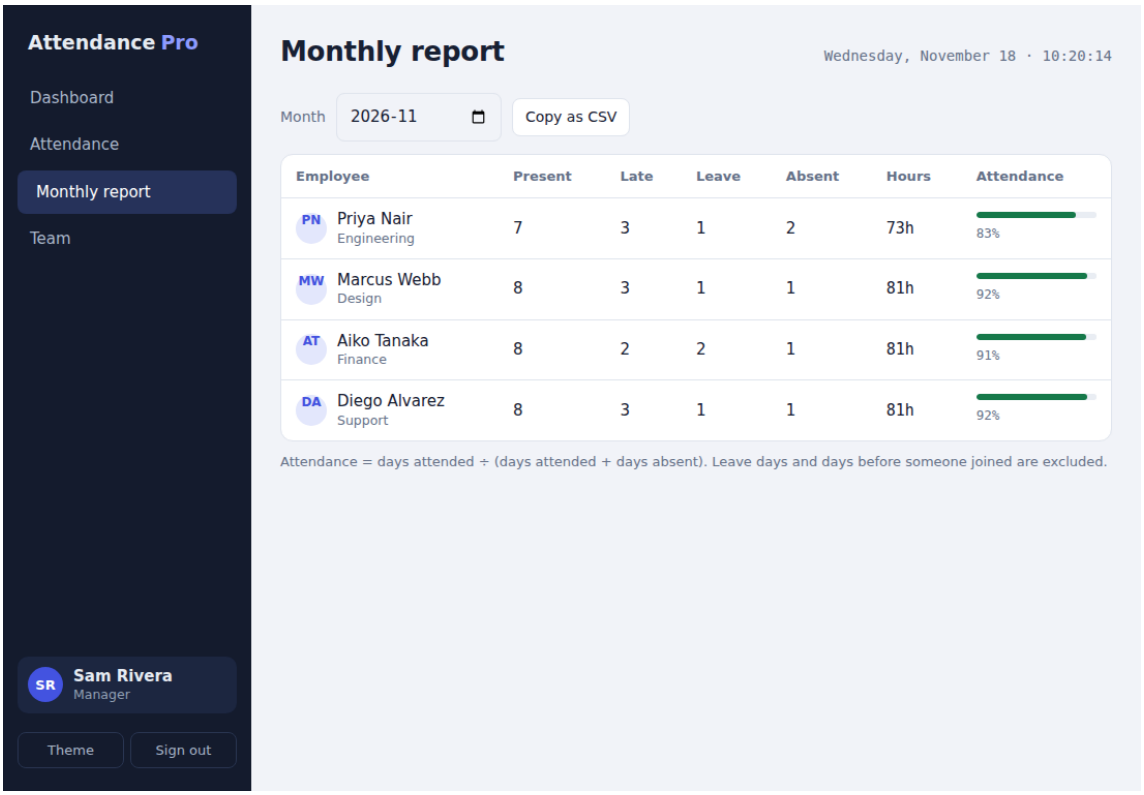


Figure 4.4: Monthly attendance report

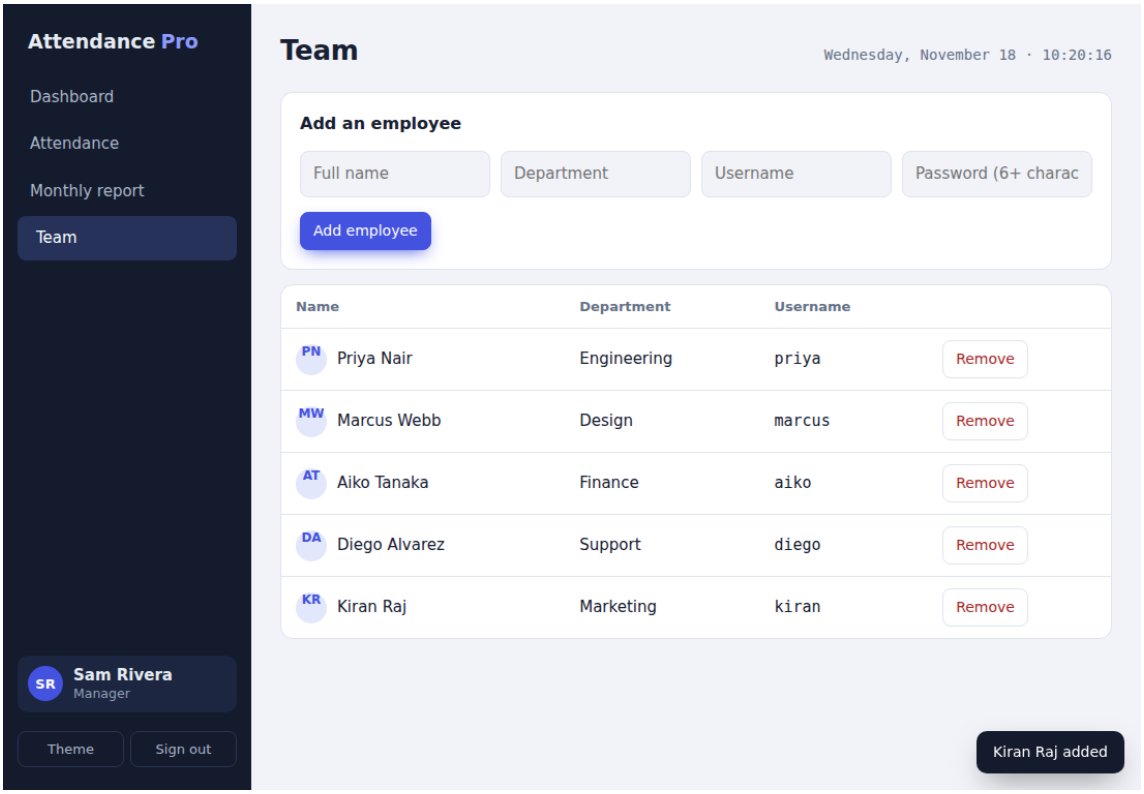


Figure 4.5: Team management

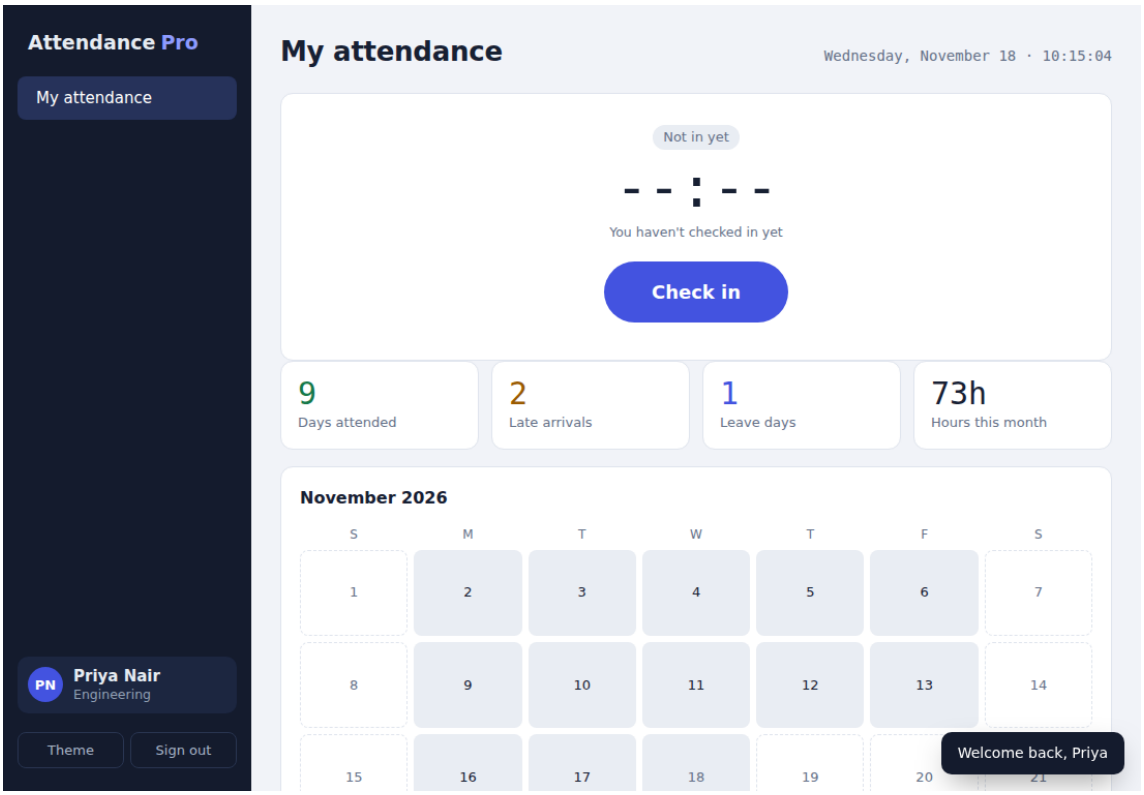


Figure 4.6: Employee check-in screen

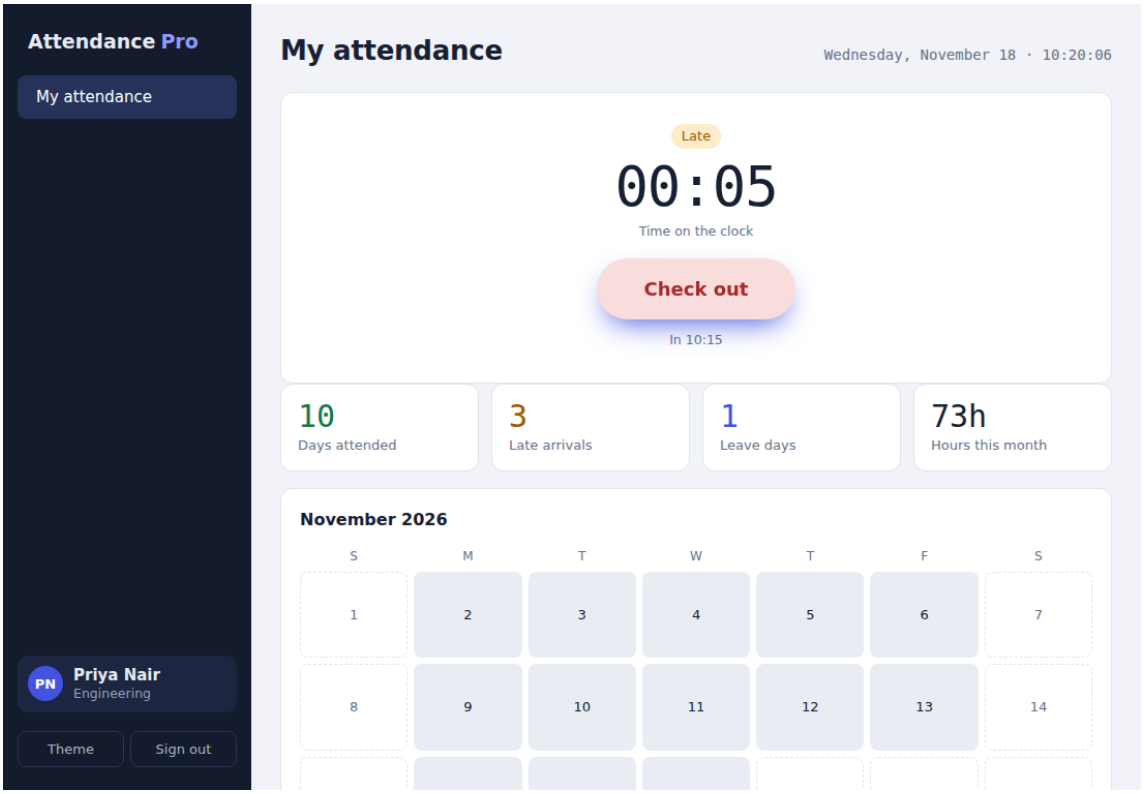


Figure 4.7: Employee checked in with live timer and calendar

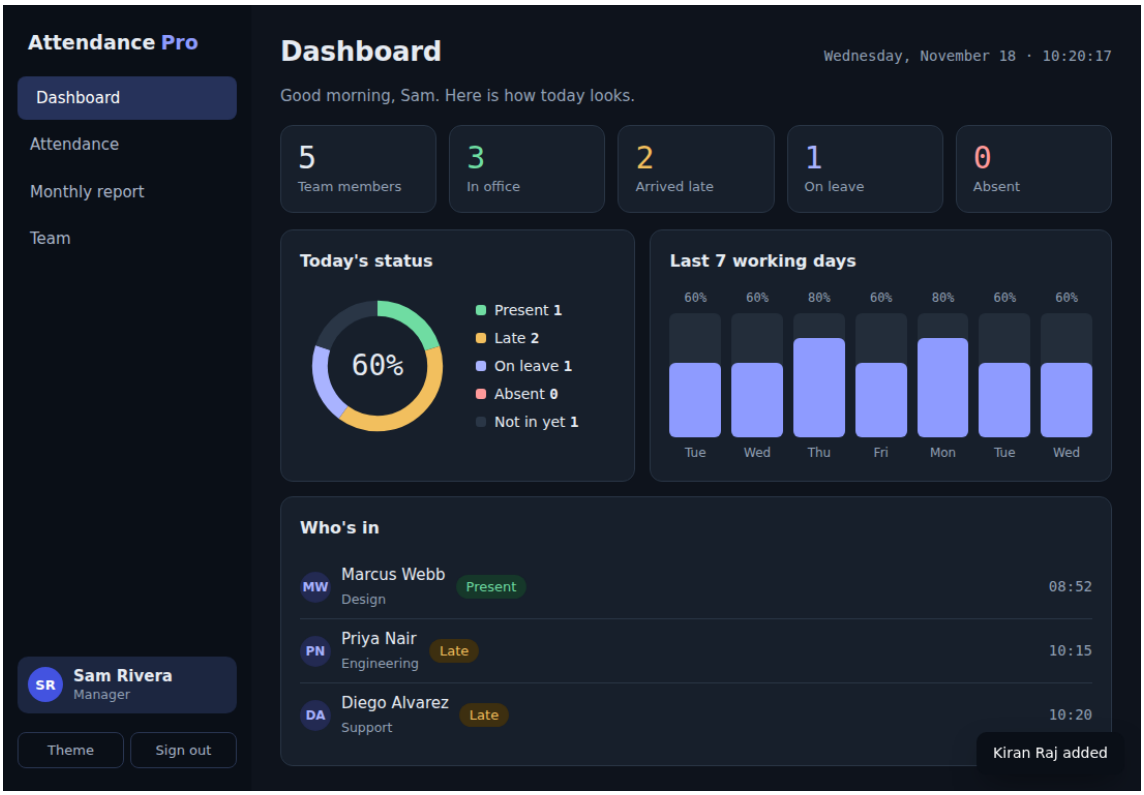


Figure 4.8: Dark theme

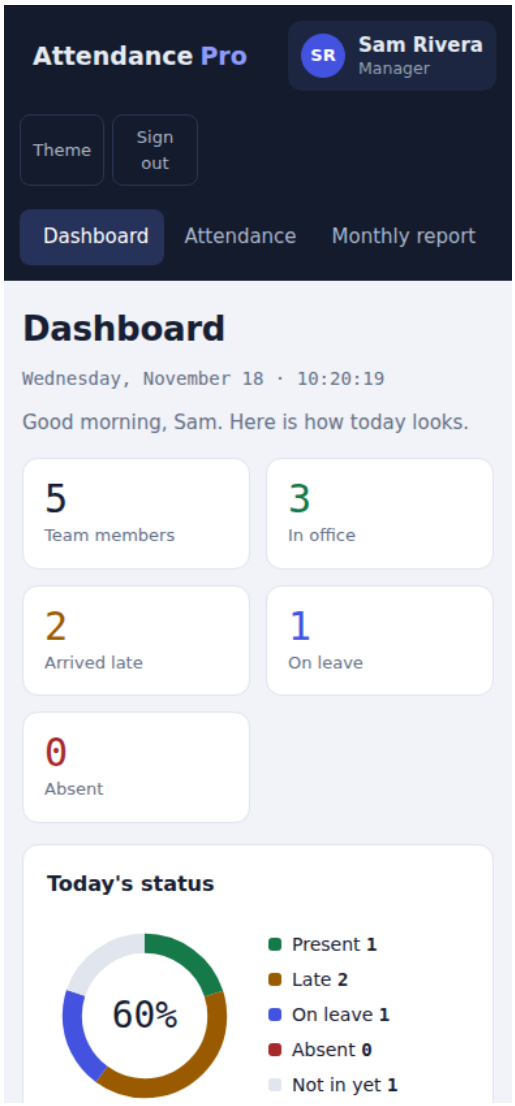


Figure 4.9: Mobile responsive view

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENT

The Employee Attendance Tracker provides a simple, accurate and accessible way to record and review attendance. Secure login, role-based screens, automatic status calculation, a dashboard and monthly reports reduce paperwork and make attendance data easy to understand. Since it runs entirely in the browser, it is fast, free to host and works on all modern devices.

Future enhancements may include a server-side database and token-based authentication so that records are shared between devices, leave requests with manager approval, QR-code, face or fingerprint check-in, location-based check-in, e-mail and push notifications, export of reports to PDF and Excel, shift and holiday calendars, integration with payroll software, and offline support as a progressive web app.

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